Exercise 2.5

1. (i) $\frac{2}{5}$ (ii) $2\sqrt{2}$ (iii) 32 (iv) 50

3. $11+6i$

8. 10 10. (i) $\pm\left(\frac{3}{\sqrt{2}}+i\frac{1}{\sqrt{2}}\right)$ (ii) $\pm\left(\sqrt{2}+i2\sqrt{2}\right)$ (iii) $\pm(2-3i)$

Exercise 2.6

3. (i) $y^2=3$ (ii) $x-y=0$ (iii) $x+y=0$ (iv) $x^2+y^2=1$

4. (i) $2+i,3$ (ii) $-1+2i,1$ (iii) $2-4i,\frac{8}{3}$

5. (i) $x^2+y^2-8x-240=0$ (ii) $6x+1=0$

Exercise 2.7

1. (i) $4\left(\cos\left(2k\pi+\frac{\pi}{3}\right)+i\sin\left(2k\pi+\frac{\pi}{3}\right)\right), k\in\mathbb{Z}$

(ii) $2\sqrt{3}\left(\cos\left(2k\pi-\frac{\pi}{6}\right)+i\sin\left(2k\pi-\frac{\pi}{6}\right)\right), k\in\mathbb{Z}$

(iii) $2\sqrt{2}\left(\cos\left(2k\pi-\frac{3\pi}{4}\right)+i\sin\left(2k\pi-\frac{3\pi}{4}\right)\right), k\in\mathbb{Z}$

(iv) $\sqrt{2}\left(\cos\left(2k\pi+\frac{5\pi}{12}\right)+i\sin\left(2k\pi+\frac{5\pi}{12}\right)\right), k\in\mathbb{Z}$

2. (i) $\frac{1}{\sqrt{2}}(1+i)$ (ii) $\frac{-i}{2}$

Exercise 2.8

3. 1 5. $3\,\mathrm{cis}\frac{\pi}{3}, -3, 3\,\mathrm{cis}\frac{5\pi}{3}$ 7. -1

9. (i) $2\sqrt{2}\,e^{\frac{i\pi}{12}}$ (ii) $2\sqrt{2}\,e^{\frac{i5\pi}{12}}$ (iii) $2\sqrt{2}\,e^{\frac{i5\pi}{4}}$

Exercise 2.9

1	2	3	4	5	6	7	8	9	10
(1)	(1)	(1)	(2)	(3)	(1)	(4)	(1)	(1)	(1)
11	12	13	14	15	16	17	18	19	20
(2)	(2)	(4)	(2)	(2)	(3)	(1)	(3)	(4)	(4)
21	22	23	24	25					
(2)	(3)	(4)	(1)	(1)					

EXERCISE 3.1

1. 60

2. (i) $x^3-6x^2+11x-6=0$ (ii) $x^3-3x+2=0$ (iii) $x^3-4x^2-4x+16=0$

3. (i) $x^3+4x^2+12x+32=0$ (ii) $4x^3+3x^2+2x+1=0$ (ii) $x^3-2x^2+3x-4=0$

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